

Reading a Rational Graph

Answer Key

Algebra 2

Quick Answers

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|---|---|
| <p>1. $\frac{x-2}{x+3}$</p> <p>2. (a) x-intercept = 4, (b) y-intercept = -4</p> <p>3. (a) simplified form = $\frac{x+2}{x+3}$, (b) vertical asymptote = -3</p> <p>4. 3</p> <p>5. (a) x-intercept = 3, (b) y-intercept = $\frac{3}{4}$</p> <p>6. (a) simplified form = $\frac{x(x+3)}{x+5}$, (b) values excluded from the domain = -5, 3</p> <p>7. (a) height of the hole = $-\frac{1}{4}$, (b) vertical asymptote = -2</p> | <p>8. $\frac{2}{5}$</p> <p>9. (a) x-intercepts = -2, 2, (b) y-intercept = -8</p> <p>10. (a) simplified form = $\frac{x+4}{x^2}$, (b) values excluded from the domain = 0, 4</p> <p>11. (a) horizontal asymptote = 2, (b) vertical asymptotes = 0, $\frac{3}{2}$, —</p> <p>12. (a) simplified form = $\frac{x(x+2)}{x+3}$, (b) x-intercepts = 0, -2, (c) height of the hole = $\frac{5}{2}$, (d) vertical asymptote = -3, —</p> |
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What is verified

23 of 25 answers machine-checked against SymPy, across 12 problems.
 — marks an answer only you can judge — problems 11, 12. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 2

HSA-APR.D.6 – problems 1, 3, 6, 10, 12

HSF-IF.C.7 – problems 2, 3, 4, 5, 7, 8, 9, 11, 12

Difficulty 1–5 of 5 across 25 tagged checks.

Common wrong answers

3. If they got -3, 3: counted the cancelled factor's zero as an asymptote as well
4. If they got 0: compared the constant terms instead of the leading coefficients and answered 0 divided by 4
5. If they got 3, -3: took every zero of the numerator as an x-intercept, including the one that cancels with the denominator
7. If they got 0: substituted $x = 2$ into the original quotient and reported the height as 0 because the numerator vanishes
8. If they got -7: used the constant terms and answered 7 over -1
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1. Simplify $\frac{x^2 - 4}{x^2 + 5x + 6}$.

Solution: $\frac{(x-2)(x+2)}{(x+2)(x+3)}$; the shared factor $(x+2)$ cancels.

$$\frac{x-2}{x+3}$$

2. $f(x) = \frac{x-4}{x+1}$. (a) x -intercept. (b) y -intercept.

Solution (a): A quotient is zero only when its numerator is zero (and the denominator is not): $x - 4 = 0$.

$$x = 4$$

Solution (b): $f(0) = \frac{0-4}{0+1}$.

$$-4$$

3. $g(x) = \frac{x^2 - x - 6}{x^2 - 9}$. (a) Simplify. (b) Vertical asymptote.

Solution (a): $\frac{(x-3)(x+2)}{(x-3)(x+3)}$, and $(x-3)$ cancels.

$$\frac{x+2}{x+3}$$

Solution (b): Only the surviving denominator factor can produce an asymptote: $x+3 = 0$. The cancelled value $x = 3$ is a hole, not an asymptote.

$$x = -3$$

4. Horizontal asymptote of $h(x) = \frac{3x^2 - x}{x^2 + 4}$.

Solution: Numerator and denominator both have degree 2, so divide every term by x^2 :

$$\frac{3x^2 - x}{x^2 + 4} = \frac{3 - \frac{1}{x}}{1 + \frac{4}{x^2}} \longrightarrow \frac{3}{1} \quad \text{as } |x| \text{ grows.}$$

$$y = 3$$

5. $u(x) = \frac{x^2 - 9}{x^2 - x - 12}$. (a) x -intercepts. (b) y -intercept.

Solution (a): $\frac{(x-3)(x+3)}{(x-4)(x+3)} = \frac{x-3}{x-4}$. The factor $(x+3)$ cancelled, so $x = -3$ is a hole, not an intercept — the graph has no point there at all. The only x -intercept comes from $x - 3 = 0$.

$$x = 3$$

Solution (b): $u(0) = \frac{0-9}{0-0-12} = \frac{-9}{-12}$.

$$\frac{3}{4}$$

6. $v(x) = \frac{x^3 - 9x}{x^2 + 2x - 15}$. (a) Simplify. (b) Excluded values.

Solution (a): $\frac{x(x-3)(x+3)}{(x+5)(x-3)}$; cancel $(x-3)$.

$$\frac{x(x+3)}{x+5}$$

Solution (b): The domain is set by the original denominator: $(x+5)(x-3) = 0$.

$$x = -5 \text{ or } x = 3$$

7. $w(x) = \frac{x^2 - 5x + 6}{x^2 - 4}$. (a) Height of the hole. (b) Vertical asymptote.

Solution (a): $\frac{(x-2)(x-3)}{(x-2)(x+2)} = \frac{x-3}{x+2}$, so the hole sits at $x = 2$. Its height is the value of the cancelled form there: $\frac{2-3}{2+2}$.

$$-\frac{1}{4}$$

Solution (b): The surviving factor is $(x+2)$.

$$x = -2$$

8. Horizontal asymptote of $r(x) = \frac{2x+7}{5x-1}$.

Solution: Equal degrees (both 1), so the ratio of the leading coefficients decides the level: $\frac{2}{5}$.

$$y = \frac{2}{5}$$

9. $s(x) = \frac{2x^2 - 8}{x^2 + 1}$. (a) x -intercepts. (b) y -intercept.

Solution (a): $2x^2 - 8 = 2(x-2)(x+2) = 0$, and the denominator $x^2 + 1$ is never zero, so nothing cancels.

$$x = -2 \text{ or } x = 2$$

Solution (b): $s(0) = \frac{-8}{1}$.

$$-8$$

10. $t(x) = \frac{x^2 - 16}{x^3 - 4x^2}$. (a) Simplify. (b) Excluded values.

Solution (a): $\frac{(x-4)(x+4)}{x^2(x-4)}$; cancel $(x-4)$.

$$\frac{x+4}{x^2}$$

Solution (b): $x^3 - 4x^2 = x^2(x-4) = 0$ gives

$$x = 0 \text{ or } x = 4$$

11. $k(x) = \frac{4x^2 + 1}{2x^2 - 3x}$. (a) Horizontal asymptote. (b) Vertical asymptotes. (c) What the degrees tell you.

Solution (a): Equal degrees, so the level is the ratio of the leading coefficients, $\frac{4}{2}$.

$$y = 2$$

Solution (b): $2x^2 - 3x = x(2x-3) = 0$, and neither factor cancels with $4x^2 + 1$.

$$x = 0 \text{ or } x = \frac{3}{2}$$

Solution (c): *Instructor-judged.* Both degrees are 2. Equal degrees mean the graph levels off, and the height it levels off at is the ratio of the *leading* coefficients, $4/2 = 2$ — the constant terms play no part. (If the top degree were smaller the level would be $y = 0$; if larger there would be no horizontal asymptote.) Full credit needs the equal-degree observation and the leading-coefficient ratio.

12. Synthesis: $F(x) = \frac{x^3 - x^2 - 6x}{x^2 - 9}$. (a) Simplify. (b) x -intercepts. (c) Height of the hole. (d) Vertical asymptote. (e) Why $x = 3$ is not an asymptote.

Solution (a): Factor both parts:

$$\frac{x(x-3)(x+2)}{(x-3)(x+3)},$$

and cancel $(x-3)$.

Solution (b): Set the simplified numerator to zero: $x(x+2) = 0$.

$$\boxed{\frac{x(x+2)}{x+3}}$$

$$\boxed{x = 0 \text{ or } x = -2}$$

Solution (c): The cancelled factor puts the hole at $x = 3$, at height $\frac{3(3+2)}{3+3} = \frac{15}{6}$.

$$\boxed{\frac{5}{2}}$$

Solution (d): The surviving denominator factor is $(x+3)$.

$$\boxed{x = -3}$$

Solution (e): *Instructor-judged.* The factor $(x-3)$ divides out of both the numerator and the denominator. After cancelling, nothing in the denominator vanishes at $x = 3$, so the values stay finite as x approaches 3 — the graph simply has one point missing there, at $(3, \frac{5}{2})$. Only $(x+3)$, which survives the cancelling, sends the graph off to infinity.